Task 1. AGM

Available marks: 12

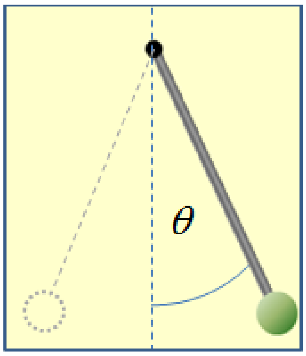
The *arithmetic-geometric mean* (AGM) of two positive real numbers is defined using an iterative algorithm. Given two numbers a_i and b_i, calculate the next pair of numbers this way:

$$a_{i+1} = \frac{(a_i + b_i)}{2} \qquad b_{i+1} = \sqrt{a_i b_i}$$

The a and b sequences converge to the same value, which is the arithmetic-geometric mean of any of the pairs, including the starting values a_0 and b_0:

$$a_\infty = b_\infty = AGM(a_0, b_0)$$

The AGM crops up in several areas of mathematics and engineering. One interesting result applies to calculating the period of an ideal frictionless pendulum.



If a pendulum of length L metres is raised to an angle θ from the vertical and then released, then if θ is a few degrees or less the period of oscillation in seconds is approximated by

$$T_{approx} = 2\pi\sqrt{L/g}$$

where g is acceleration due to gravity in metres per second per second.

An equation, discovered only in 2012, describes the exact period for any angle up to 180 degrees (where the pendulum starts in an upside down position):

$$T_{exact} = \frac{T_{approx}}{AGM\left(1, \cos\frac{\theta}{2}\right)}$$

Your Task

The task has two parts. First write a program that calculates the AGM of two positive values, accurate to 10 decimal places. That is, you can stop the iteration when the current values differ by less than 0.5e–10 or 5e–11.

Test data in **task1partA.dat** consists of a number of lines, each containing a pair of values. Display the numbers and the calculated AGM to the required 10-digit precision. This part is worth **5 marks**.

Part B over...

Part B

For a further **4 marks**, extend your program so that it calculates and displays the pendulum starting angle that corresponds to a given ratio

$$R = T_{exact} \,\Big/\, T_{approx}$$

The angle should be expressed in degrees, accurate to 2 decimal places. Your program should track and accurately report the number of AGM evaluations required to obtain the answer. The last **3 marks** is awarded if your program performs an intelligent search such as bisection* and the number of AGM evaluations is appropriate (the limit is not revealed here, but the judges will indicate whether or not your solution is sufficiently efficient).

Test cases in **task1partB.dat** are single values for R, each greater than 1 and less than 10.

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* Bisection is a strategy that maintains lower and upper bounds that converge to a single solution, by evaluating the midpoint to decide which bound to move on each iteration. Binary search in an ordered array is an example of bisection.